9th Class • Punjab Board • PECTAA New Syllabus

Chapter 5: Web Development with HTML, CSS and JavaScript

Complete solved exercise — HTML, CSS, JavaScript, responsive design and runnable code examples.

Chapter 5 • Complete official exercise • 8 MCQs + 9 short + 5 long questions

What this chapter is about

• **HTML** supplies page structure and content, **CSS** controls presentation and layout, and **JavaScript** adds behaviour and interactivity.
• A web document needs valid elements and attributes. CSS can be inline, internal or external; JavaScript can also be embedded or loaded from an external file.
• The CSS box model consists of content, padding, border and margin. Responsive pages adapt to different screen sizes, normally with a viewport meta tag, flexible CSS and media queries; JavaScript can enhance this behaviour.
• Browser developer tools help inspect HTML/CSS, read JavaScript errors, test output and debug code.

Source-check notes for this chapter

• **MCQ 1:** the printed key selects <footer>, so the expected answer is shown, but the item is technically outdated because <footer> is valid HTML5.
• **MCQ 6:** the printed key selects (b) /--/, which is not HTML comment syntax. The verified intended option is (d) <!--, and a complete comment is <!-- comment -->.

Coverage: Chapter 5 corresponds to Unit 8 Web Development with HTML, CSS, and JavaScript in the twelve-unit textbook. The combined-edition exercise order is retained.

A. Multiple Choice Questions

1. Which of the following tag is not a correct HTML tag?

(a) (b)
(c) <head> (d) <footer>

Answer: (d) <footer> — **Textbook key: (d).** Source caution: modern HTML5 does define <footer> as a valid semantic element, so the question is outdated. The expected examination option is retained, but students should not learn that <footer> is invalid HTML5.

2. What does CSS stand for?

(a) Cascading Style Sheets (b) Computer Style Sheets
(c) Creative Style Sheets (d) Colorful Style Sheets

Answer: (a) Cascading Style Sheets — CSS means Cascading Style Sheets and controls presentation such as colours, spacing and layout.

3. Which of the following tag is used to create a hyperlink in HTML?

(a) <link> (b) <a>
(c) <href> (d) <nav>

Answer: (b) <a> — The anchor element <a> creates a hyperlink; href is its destination attribute, not a tag.

4. Which HTML attribute is used to define inline styles?

(a) class (b) style
(c) font (d) styles

Answer: (b) style — The style attribute places CSS declarations directly on one element, for example <p style="color:red">.

5. Which of the following is the correct syntax for a CSS rule?

(a) selector {property: value;} (b) selector: {property=value;}
(c) selector {property=value} (d) selector: {property: value;}

Answer: (a) selector {property: value;} — A CSS rule has a selector followed by a declaration block; each property and value are separated by a colon.

6. In HTML, which markup is used for comments?

(a) /**/ (b) /--/
(c) <-- (d) <!--

Answer: (d) <!-- — Option (d) is the correct **opening** delimiter. A complete comment is **<!-- comment -->**. **Source correction:** the printed answer table says (b), but /--/ is not an HTML comment; this is a print/key error.

7. Which HTML tag is used to create an unordered list?

(a) (b)
(c) (d) <list>

**Answer: (b) ** — The <ul> element creates a bulleted list, with each item inside <li>.

8. Which tag is used to display a horizontal line in HTML?

(a)
 (b) <hr>
(c) <line> (d) <hline>

Answer: (b) <hr> — The <hr> element represents a thematic break, normally displayed as a horizontal rule;
 creates a line break.

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B. Short Questions

1. What is the purpose of the <head> tag in HTML?

The <head> element contains metadata and supporting information for the document rather than its visible body content. It commonly holds the page <title>, character-set and viewport <meta> tags, and links to CSS or script resources.

2. Explain the difference between an ordered list and an unordered list in HTML.

Ordered list	Unordered list
Created with .	Created with .
Items are normally numbered or lettered, so sequence matters.	Items are normally bulleted, so no particular order is implied.
Example use: steps in a recipe or installation instructions.	Example use: a shopping list or a set of features.
Each item still uses .	Each item also uses .

3. How do you add a comment in HTML?

Place the comment between **<!--** and **-->**, for example:
<!-- This section contains the navigation links -->
The browser ignores the comment when rendering the page. Both the opening and closing delimiters are required.

4. What are the different ways to apply CSS to an HTML document?

Method	Form	Best use
Inline CSS	A style attribute on one element: <p style="color: blue;">.	A single special case; generally avoided for large sites because it mixes structure and presentation.
Internal CSS	A <style>…</style> block inside the document's <head>.	Rules needed only by one HTML page.
External CSS	A separate .css file linked with <link rel="stylesheet" href="style.css">.	Preferred for consistent styling and reuse across many pages.

5. How can you include JavaScript in an HTML file?

JavaScript can be placed directly in a <script> element or loaded from a separate .js file. External code is easier to reuse and maintain.

Two inclusion methods

```
<!-- Internal JavaScript -->
<script>
  alert("Welcome");
</script>

<!-- External JavaScript -->
<script src="script.js"></script>
```

The script is commonly placed before the closing </body> tag so that the HTML elements are available. In modern HTML, an external script in the <head> can also use the defer attribute: <script src="script.js" defer></script>.

6. Describe the syntax for creating a hyperlink in HTML.

Basic hyperlink

```
<a href="https://www.example.com">Visit Example</a>
```

The <a> element is the anchor, **href** gives the destination URL and the text between the opening and closing tags is what the user selects. A relative link such as About us opens another file in the same site.

7. What is the function of the tag in HTML?

The element is a generic block-level container used to group related content for layout or styling. It has no special semantic meaning by itself; meaningful elements such as <header>, <main> or <section> should be used where appropriate.

8. What is the use of the <table> tag in HTML?

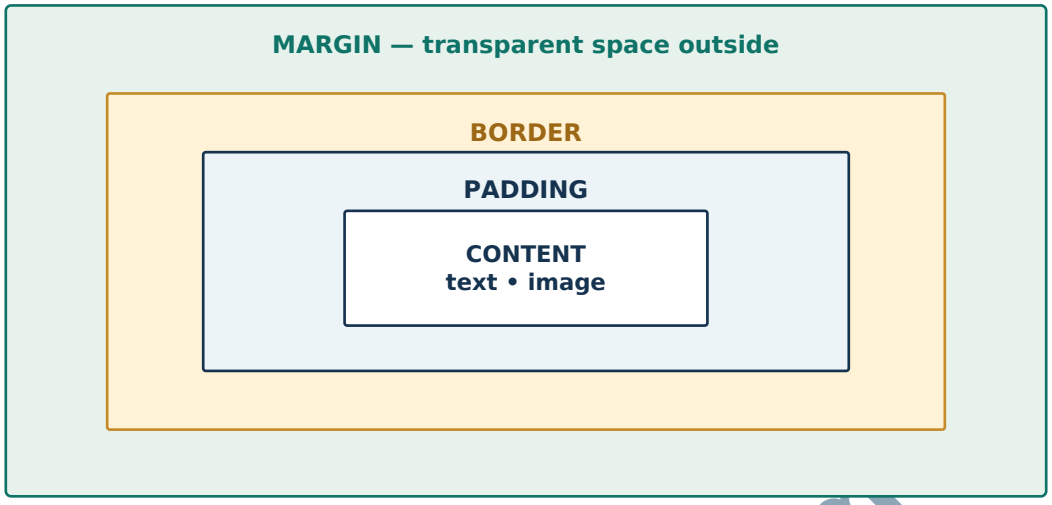
The <table> element presents data in rows and columns. A row uses <tr>, a header cell uses <th> and an ordinary data cell uses <td>. Tables are for tabular information, not for arranging an entire webpage layout.

9. Explain the box model in CSS.

Every ordinary HTML element can be imagined as a rectangular box. From the centre outwards, it contains **content → padding → border → margin**.

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Figure 1 — Content, padding, border and margin



• **Content:** the text, image or other material, with a defined content width and height.

• **Padding:** transparent space between content and border; it increases the inner breathing room.

• **Border:** the visible line surrounding padding and content.

• **Margin:** transparent space outside the border that separates the element from neighbouring elements.

With the browser's default content-box sizing, total occupied width = left/right content width + padding + border + margin. CSS box-sizing: border-box makes the declared width include content, padding and border, but not margin.

C. Long Questions

1. Explain the process of setting up a development environment for web development. By disscussing the necessary softwares and tools.

1. Create a project folder: Make a clear folder such as school-site and organise subfolders for images, CSS and JavaScript.

2. Install a code editor: Visual Studio Code is a common choice. Useful features include syntax highlighting, auto-completion, file search and an integrated terminal.

3. Install modern browsers: Use a current browser such as Chrome, Firefox or Edge. Testing in more than one browser reveals compatibility differences.

4. Create the starting files: Add index.html and, when required, style.css and script.js. Link the CSS and JavaScript correctly and use relative paths for local images.

5. Use developer tools: Open the browser's Elements/Inspector panel for HTML/CSS and Console for JavaScript errors. The Network and responsive-design tools help check loading and screen sizes.

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6. Run the page: A simple HTML/CSS/JavaScript page can open directly in a browser. A local development server is useful for projects that require HTTP features or a framework.

7. Add supporting tools as needed: Git records versions and supports backup/collaboration; Node.js may provide a local server and packages; image editors optimise web graphics. These are not all necessary for a first static page.

8. Test and validate: Check links, image alt text, console errors, keyboard use and mobile layout before publishing. Keep a backup of working versions.

For this syllabus, the essential minimum is a **folder + text/code editor + modern web browser + developer tools**. Other software supports larger projects but should not hide the basic HTML, CSS and JavaScript workflow.

2. Create a basic HTML page that includes a header, a paragraph, an image, and a hyperlink.

Complete index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>My School Page</title>
</head>
<body>
  <header>
    <h1>Welcome to My School</h1>
  </header>

  <main>
    <p>We learn, share ideas and work together.</p>
    <img src="school.jpg" alt="The front of our school building" width="400">
    <p>
      <a href="https://www.example.com">Visit our learning website</a>
    </p>
  </main>
</body>
</html>
```

Place an image named **school.jpg** in the same folder as index.html, or change src to its correct path. The <header> contains the heading, <p> supplies the paragraph, <img> supplies the image with alternative text and <a href> creates the hyperlink. The alt text describes the image if it cannot load and for assistive technology.

3. Create a table using HTML which contains 3 rows and 2 column?

Exactly three rows and two cells in each row

```
<table border="1">
  <tr>
    <th>Student</th>
    <th>Marks</th>
  </tr>
  <tr>
    <td>Ali</td>
    <td>85</td>
  </tr>
  <tr>
    <td>Sara</td>
    <td>92</td>
  </tr>
</table>
```

The first <tr> is the header row and the next two are data rows. Each has two cells, giving two columns. In a finished page, CSS border and padding rules are preferable to the presentational border attribute.

4. Describe any three CSS selectors and provide examples of each.

Selector	Selects	Example
Type / element selector	Every element of the stated HTML type.	p { color: navy; } styles all <p> elements.
Class selector	Elements whose class attribute contains the chosen name; reusable on many elements.	.notice { background: lightyellow; } used as class="notice".
ID selector	The one element with a unique id value in the document.	#main-title { text-align: center; } used as id="main-title".

CSS

```
/* Type selector */
p { line-height: 1.5; }

/* Class selector: reusable */
.notice { background-color: lightyellow; }

/* ID selector: unique on this page */
#main-title { text-align: center; }
```

Matching HTML

```
<h1 id="main-title">Class 9 Web Page</h1>
<p class="notice">Registration closes on Friday.</p>
<p>This is an ordinary paragraph.</p>
```

A class begins with a dot and an ID with a hash in CSS. An ID must be unique within the document; the same class may be assigned to many elements.

5. Explain the process of creating a responsive web page using java script with the help of an examples.

A responsive page adapts its layout to the screen instead of forcing one fixed desktop design. The foundation is a viewport meta tag, flexible sizes and readable content. CSS media queries normally handle layout breakpoints; **JavaScript can detect viewport changes and enhance the layout or controls**, as requested here.

Responsive HTML, CSS and JavaScript example

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Responsive Page</title>
  <style>
    .container { width: 80%; margin: auto; font-size: 18px; }
    .compact .container { width: 96%; font-size: 15px; }
    img { max-width: 100%; height: auto; }
  </style>
</head>
<body>
  <main class="container">
    <h1>Computer Science</h1>
    <p>This text and layout adapt to the viewport.</p>
  </main>

  <script>
    function makeResponsive() {
      const smallScreen = window.innerWidth < 600;
      document.body.classList.toggle("compact", smallScreen);
    }

    window.addEventListener("resize", makeResponsive);
    makeResponsive();  // Run once when the page loads
  </script>
</body>
</html>
```

1. The viewport meta tag makes the CSS viewport match the device's width rather than a default scaled desktop width.

2. The normal .container is flexible because it uses a percentage width. The image also shrinks within its container.

3. makeResponsive() checks window.innerWidth. Below 600 pixels, it adds the compact class to the body; on a larger screen, it removes it.

4. The resize listener calls the function whenever the viewport changes. Calling the function once at load applies the correct initial state.

5. Test by resizing the browser or using responsive developer tools. Also test real touch controls, readable text and image loading.

For most real projects, prefer a CSS media query such as **@media (max-width: 600px)** for pure layout and reserve JavaScript for interactive behaviour such as

opening a mobile navigation menu.

D. More Important Questions (from the text)

1. Differentiate between HTML, CSS and JavaScript.

HTML	CSS	JavaScript
HyperText Markup Language.	Cascading Style Sheets.	A programming language for web behaviour.
Defines structure, content and meaning.	Defines presentation, layout and visual effects.	Defines actions, decisions and dynamic changes.
Examples: headings, paragraphs, images, links and tables.	Examples: colours, fonts, spacing, selectors and media queries.	Examples: button responses, validation and changing styles.
Question answered: What is on the page?	Question answered: How should it look?	Question answered: What should it do?

2. How do you link an external CSS file and debug a webpage?

External resources in the document head

```
<head>
  <link rel="stylesheet" href="style.css">
  <script src="script.js" defer></script>
</head>
```

For debugging, reproduce the problem, inspect the HTML structure and computed CSS in browser developer tools, and read Console errors from the first error downwards. Check file names and paths, missing brackets/tags, selector specificity and JavaScript variable values. Make one small correction at a time and retest. CSS comments use /* comment */; JavaScript single-line comments use // and HTML comments use <!-- comment -->.

3. Official-unit extension: How do CSS transitions and animations improve a webpage?

A **transition** smoothly changes a property between two states, such as normal and hover. An **animation** uses @keyframes to define one or more stages and can repeat without requiring hover.

Transition and keyframe animation examples

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```
button {
  background-color: navy;
  color: white;
  transition: background-color 0.4s ease;
}
button:hover { background-color: teal; }

@keyframes pulse {
  from { opacity: 0.5; }
  to   { opacity: 1; }
}
.notice { animation: pulse 1s ease-in-out 3 alternate; }
```

Use motion sparingly so it supports understanding rather than distracting the user. Respect reduced-motion preferences and avoid making essential information depend only on animation.

4. Official-unit extension: How can JavaScript change webpage text when a button is clicked?

Runnable button-event example

```
<p id="message">Hello!</p>
<button type="button" id="changeButton">Change text</button>

<script>
  const button = document.getElementById("changeButton");
  button.addEventListener("click", function () {
    document.getElementById("message").textContent = "Welcome to my page!";
  });
</script>
```

getElementById() selects an element by its unique ID. addEventListener() runs the function only when the button is clicked, and textContent safely replaces the paragraph's text. Place the script after the elements, as shown, or load it with defer so the elements exist before the script runs.

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Points to Remember

HTML: structure/content; **CSS:** presentation; **JavaScript:** behaviour and interactivity.

Hyperlink: link text. **Unordered list:** with <li> items; ordered list uses <ol>.

HTML comment: <!-- comment -->. **CSS comment:** /* comment */. **JavaScript line comment:** // comment.

CSS rule: selector { property: value; }. Type selectors use a name, classes a dot and IDs a hash.

Box model: content → padding → border → margin. Padding is inside the border; margin is outside it.

Responsive design: viewport meta tag, flexible dimensions/media and suitable breakpoints; JavaScript can add adaptive behaviour.

Modern source caution: <footer> is valid HTML5 even though the printed MCQ expects it as the invalid tag.

Debugging: inspect elements/styles, read Console errors, check paths and syntax, change one thing and retest.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	d	2	a	3	b	4	b	5	a
6	d*	7	b	8	b				

* Source correction: Q6: printed table = (b). See the reason under the affected MCQ for the verified answer.

How to attempt this chapter in the paper

- Write code with matching opening/closing tags and quote attribute values. Use < and > only as text entities when showing markup inside another HTML text example.
- For an HTML-table question, count <tr> rows and cells carefully; Q3 requires exactly three rows and two columns.
- For a selector answer, show both the CSS rule and the HTML to which it applies.
- In responsive answers, include the viewport meta tag and test small and large widths; do not claim JavaScript alone replaces responsive CSS.

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